In [1]:
```
import numpy as np
arr=np.array([100,200,300,400])
arr
```

Out[1]:
```
array([100, 200, 300, 400])
```

In [2]:
```
print("Array:", arr)
print("Dimensions:", arr.ndim)
print("Shape:", arr.shape)
print("Size:", arr.size)
```

```
Array: [100 200 300 400]
Dimensions: 1
Shape: (4,)
Size: 4
```

In [3]:
```
print("Total:", np.sum(arr))
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))
print("Mean:", np.mean(arr))
```

```
Total: 1000
Maximum: 400
Minimum: 100
Mean: 250.0
```

In [4]:
```
new_increase=arr*1.10
print(new_increase)
```

```
[110. 220. 330. 440.]
```

In [7]:
```
data=arr[arr>200]
mark_data=arr>200
print(mark_data)
print(data)
```

```
[False False  True  True]
[300 400]
```

In [1]:
```
import numpy as np
import pandas as pd

df=pd.read_csv("waste_management.csv")
print(df)
```

```
  Bin_ID     Zone  Daily_Waste_KG  Recycled_Waste_KG  Bin_Fill_Percent  \
0   B201    North             120                 45                75
1   B202    South             180                 60                90
2   B203     East              95                 50                55
3   B204     West             220                 70                98
4   B205  Central             150                 80                80
5   B206    North             140                 65                70
6   B207    South              85                 40                45
7   B208     East             200                 55                95
8   B209     West             110                 60                60
9   B210  Central             170                 90                85

   Collection_Delay_Minutes
0                        10
1                        25
2                         0
3                        35
4                        15
5                         5
6                         0
7                        30
8                         8
9                        20
```

In [2]:
```
print("Dimensions:", df.ndim)
print("Shape:", df.shape)
print("Size:", df.size)

print('Info:',df.info())
print('Describe:',df.describe())
print('Is Null:',df.isnull())
print('Is Null Count:',df.isnull().sum())
```

```
Dimensions: 2
Shape: (10, 6)
Size: 60
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Bin_ID                    10 non-null     str
 1   Zone                      10 non-null     str
 2   Daily_Waste_KG            10 non-null     int64
 3   Recycled_Waste_KG         10 non-null     int64
 4   Bin_Fill_Percent          10 non-null     int64
 5   Collection_Delay_Minutes  10 non-null     int64
dtypes: int64(4), str(2)
memory usage: 612.0 bytes
Info: None
Describe:        Daily_Waste_KG  Recycled_Waste_KG  Bin_Fill_Percent  \
count       10.000000          10.000000         10.000000
mean       147.000000          61.500000         75.300000
std         45.289194          15.465014         17.726628
min         85.000000          40.000000         45.000000
25%        112.500000          51.250000         62.500000
50%        145.000000          60.000000         77.500000
75%        177.500000          68.750000         88.750000
max        220.000000          90.000000         98.000000

       Collection_Delay_Minutes
count                 10.000000
mean                  14.800000
std                   12.354037
min                    0.000000
25%                    5.750000
50%                   12.500000
75%                   23.750000
max                   35.000000
Is Null:    Bin_ID   Zone  Daily_Waste_KG  Recycled_Waste_KG
Bin_Fill_Percent  \
0   False  False           False              False             False
1   False  False           False              False             False
2   False  False           False              False             False
3   False  False           False              False             False
4   False  False           False              False             False
5   False  False           False              False             False
```

```
6   False  False           False              False             False
7   False  False           False              False             False
8   False  False           False              False             False
9   False  False           False              False             False

   Collection_Delay_Minutes
0                     False
1                     False
2                     False
3                     False
4                     False
5                     False
6                     False
7                     False
8                     False
9                     False
Is Null Count: Bin_ID                      0
Zone                         0
Daily_Waste_KG               0
Recycled_Waste_KG            0
Bin_Fill_Percent             0
Collection_Delay_Minutes     0
dtype: int64
```

In [3]:
```
print("Total waste collected:", df["Daily_Waste_KG"].sum())
print("Maximum waste:", df["Daily_Waste_KG"].max())
print("Minimum waste:", df["Daily_Waste_KG"].min())
print("Average waste:", df["Daily_Waste_KG"].mean())
```

```
Total waste collected: 1470
Maximum waste: 220
Minimum waste: 85
Average waste: 147.0
```

In [4]:
```
df["Daily_Waste_KG"].to_numpy()
```

Out[4]:
```
array([120, 180,  95, 220, 150, 140,  85, 200, 110, 170])
```

In [5]:
```
print(np.sum(df["Daily_Waste_KG"].to_numpy(), axis=0))
```

```
1470
```

In [6]:
```
df.head()
```

Out[6]:

	Bin_ID	Zone	Daily_Waste_KG	Recycled_Waste_KG	Bin_Fill_Percent	Collection_Delay_Minutes
0	B201	North	120	45	75	10
1	B202	South	180	60	90	25
2	B203	East	95	50	55	0
3	B204	West	220	70	98	35
4	B205	Central	150	80	80	15

In [7]:
```
df.tail(4)
```

Out[7]:

	Bin_ID	Zone	Daily_Waste_KG	Recycled_Waste_KG	Bin_Fill_Percent	Collection_Delay_Minutes
6	B207	South	85	40	45	0
7	B208	East	200	55	95	30
8	B209	West	110	60	60	8
9	B210	Central	170	90	85	20

In [8]:
```
print(df.columns.tolist())
```

```
['Bin_ID', 'Zone', 'Daily_Waste_KG', 'Recycled_Waste_KG',
'Bin_Fill_Percent', 'Collection_Delay_Minutes']
```

In [9]:
```
print(df.info())
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 6 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Bin_ID                   10 non-null     str
 1   Zone                     10 non-null     str
 2   Daily_Waste_KG           10 non-null     int64
 3   Recycled_Waste_KG        10 non-null     int64
 4   Bin_Fill_Percent         10 non-null     int64
 5   Collection_Delay_Minutes  10 non-null     int64
dtypes: int64(4), str(2)
memory usage: 612.0 bytes
None
```

In [10]:
```
df["Increased_Waste"] = df["Daily_Waste_KG"] * 1.10

df[["Bin_ID", "Daily_Waste_KG", "Increased_Waste"]]
```

Out[10]:

	Bin_ID	Daily_Waste_KG	Increased_Waste
0	B201	120	132.0
1	B202	180	198.0
2	B203	95	104.5
3	B204	220	242.0
4	B205	150	165.0
5	B206	140	154.0
6	B207	85	93.5
7	B208	200	220.0
8	B209	110	121.0
9	B210	170	187.0

In [11]:
```
high_waste = df[df["Daily_Waste_KG"] > 150]
high_waste
```

Out[11]:

	Bin_ID	Zone	Daily_Waste_KG	Recycled_Waste_KG	Bin_Fill_Percent	Collection_Delay_Minutes	Increased_Waste
1	B202	South	180	60	90	25	198.0
3	B204	West	220	70	98	35	242.0
7	B208	East	200	55	95	30	220.0
9	B210	Central	170	90	85	20	187.0

In [12]:
```
high_fill = df[df["Bin_Fill_Percent"] > 80]
high_fill
```

Out[12]:

	Bin_ID	Zone	Daily_Waste_KG	Recycled_Waste_KG	Bin_Fill_Percent	Collection_Delay_Minutes	Increased_Waste
1	B202	South	180	60	90	25	198.0
3	B204	West	220	70	98	35	242.0
7	B208	East	200	55	95	30	220.0
9	B210	Central	170	90	85	20	187.0

In [13]:
```
df[(df["Daily_Waste_KG"] > 120) & (df["Recycled_Waste_KG"] < 70)]
```

Out[13]:

	Bin_ID	Zone	Daily_Waste_KG	Recycled_Waste_KG	Bin_Fill_Percent	Collection_Delay_Minutes	Increased_Waste
1	B202	South	180	60	90	25	198.0
5	B206	North	140	65	70	5	154.0
7	B208	East	200	55	95	30	220.0

In [14]:
```
df["Non_Recycled_Waste_KG"] = (
    df["Daily_Waste_KG"] - df["Recycled_Waste_KG"]
)

df[["Bin_ID", "Daily_Waste_KG",
    "Recycled_Waste_KG", "Non_Recycled_Waste_KG"]]
```

Out[14]:

	Bin_ID	Daily_Waste_KG	Recycled_Waste_KG	Non_Recycled_Waste_KG
0	B201	120	45	75
1	B202	180	60	120
2	B203	95	50	45
3	B204	220	70	150
4	B205	150	80	70
5	B206	140	65	75
6	B207	85	40	45
7	B208	200	55	145
8	B209	110	60	50
9	B210	170	90	80

In [15]:
```
non_recycled = df["Non_Recycled_Waste_KG"]

print("Total non-recycled waste:", non_recycled.sum())
print("Average non-recycled waste:", non_recycled.mean())
print("Minimum non-recycled waste:", non_recycled.min())
print("Maximum non-recycled waste:", non_recycled.max())
```

```
Total non-recycled waste: 855
Average non-recycled waste: 85.5
Minimum non-recycled waste: 45
Maximum non-recycled waste: 150
```